Soil Moisture Forecasting for Precision Agriculture

FULL PROJECT SYNOPSIS

Project Overview

Soil Moisture Forecasting for Precision Agriculture combines historical agricultural sensor data, machine-learning forecasts, a Streamlit dashboard, and an ESP32/Wokwi demonstration to support irrigation planning.

Data and Forecasting

Python scripts clean historical farm records, prepare time-based features, train forecasting models, and evaluate saved predictions at 1-, 3-, and 6-hour horizons. The dashboard loads existing trained artifacts; it does not retrain models at startup. Wokwi readings are collected separately and are not mixed with historical training data.

Dashboard

The six-page Streamlit dashboard presents a farm overview, soil-moisture predictions, visualizations, model-performance comparisons, irrigation recommendations, and recorded Wokwi data. Recommendations are decision support: the dashboard does not directly control the simulator or irrigation hardware.

ESP32 and Wokwi Demonstration

The ESP32 reads or simulates soil moisture, temperature, and air humidity. At firmware defaults, moisture below 30% activates the relay and red dry-condition LED; moisture above 70% turns on the yellow wet-condition LED. The circuit is described in diagram.json. A Python serial collector stores simulator readings in a separate CSV.

Key Files

Firmware: src/main.cpp | Circuit: diagram.json | Dashboard: dashboard/app.py | Collector: backend/collect_hardware_data.py | Model training: backend/train_model.py | Model evaluation: backend/evaluate_model.py

Forecasts and simulated readings should be interpreted with local agronomic knowledge.